

Customer Churn Analysis

Analytics Report — Retention Metrics & Behavioral Drivers

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46.67%

Churn Rate

Rs. 1,591.55

Est. Customer LTV

Rs. 409.93

MRR Lost / Month

15

Customers Analyzed

1. Executive Summary

This report analyzes a customer subscription dataset of 15 accounts to quantify churn, estimate revenue impact, and identify the behavioral and contractual factors most strongly associated with cancellations. The objective is to support data-driven retention decisions.

The overall churn rate stands at **46.67%** (7 of 15 customers), translating to a **Rs. 409.93 loss in Monthly Recurring Revenue (MRR)** — roughly **32.3%** of total MRR. The estimated **Customer Lifetime Value (CLV)** is **Rs. 1,591.55** per customer, based on average monthly revenue and average subscription tenure.

Three factors emerge as the strongest churn drivers: **Month-to-Month contracts** (100% churn), **high support-ticket volume** (customers with 3+ tickets churn at 100%), and **short account tenure** (churned customers average just 7 months vs. 29.1 months for retained customers). These findings point to clear, actionable retention levers detailed in Section 6.

2. Dataset Overview

The dataset contains 15 customer records with 11 attributes covering demographics, subscription details, billing, support interactions, and churn status. No missing values or duplicate rows were found during data cleaning.

Attribute	Details
Total Customers	15
Features	Gender, Age, Tenure (Months), Subscription Type, Monthly Charges, Total Charges, Contract Type, Support Tickets, Payment Method, Churn
Missing Values	0 (across all columns)
Duplicate Rows	0
Subscription Tiers	Basic, Pro, Enterprise
Contract Types	Month-to-Month, One Year, Two Year
Payment Methods	Credit Card, Debit Card, Bank Transfer, UPI

3. Core Retention Metrics

3.1 Baseline Churn Rate

Formula: $\text{Churn Rate} = (\text{Churned Customers} / \text{Total Customers}) \times 100$

Churned Customers: 7 | Total Customers: 15 | Churn Rate = 46.67%

3.2 Monthly Recurring Revenue (MRR) Loss

Formula: $\text{MRR Loss} = \text{Sum of Monthly Charges for churned customers}$

Metric	Value (Rs.)
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Total MRR (all customers)	1,269.85
MRR from retained customers	859.92
MRR Lost (churned customers)	409.93
% of MRR at risk	32.3%

3.3 Customer Lifetime Value (CLV)

Formula: CLV = Average Monthly Revenue x Average Customer Lifetime (months)

Metric	Value
Average Monthly Revenue	Rs. 84.66
Average Customer Lifetime (Tenure)	18.8 months
Estimated CLV	Rs. 1,591.55

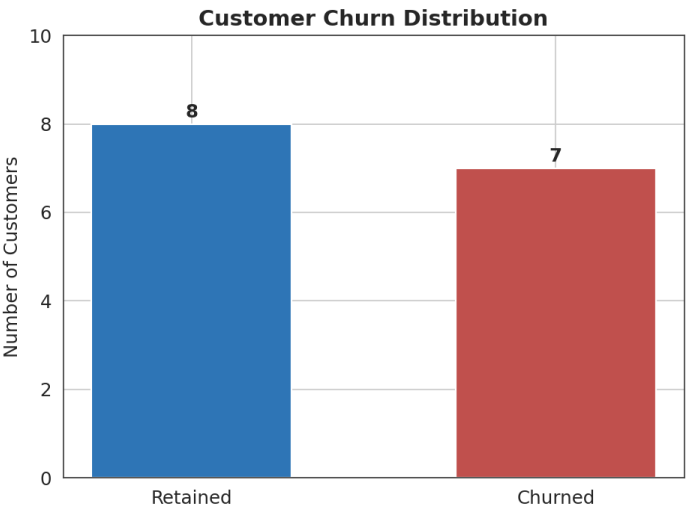


Figure 1: Customer churn distribution — retained vs. churned customers.

4. Churn Rate by Segment

Churn was grouped by contract type, tenure, and payment method to identify which customer segments cancel most frequently.

4.1 By Contract Type

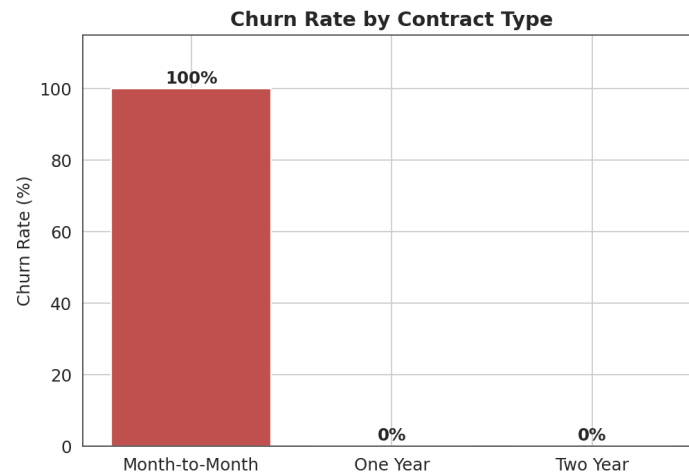


Figure 2: Churn rate by contract type.

Month-to-Month subscribers churn at **100%**, while every One Year and Two Year customer in the dataset was retained (0% churn). Contract length is the single strongest predictor of churn.

4.2 By Payment Method

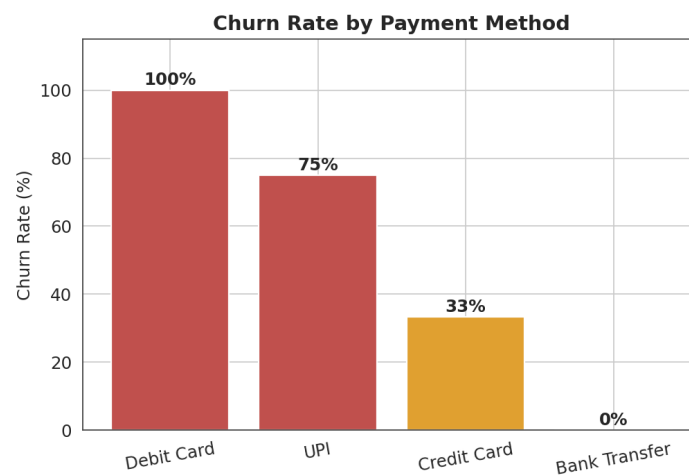


Figure 3: Churn rate by payment method.

Debit Card users churn at **100%** and UPI users at **75%**, compared to 33.3% for Credit Card and 0% for Bank Transfer. Manual, low-friction payment methods correlate with higher churn, possibly reflecting weaker payment commitment or failed renewals.

4.3 By Tenure (Months)

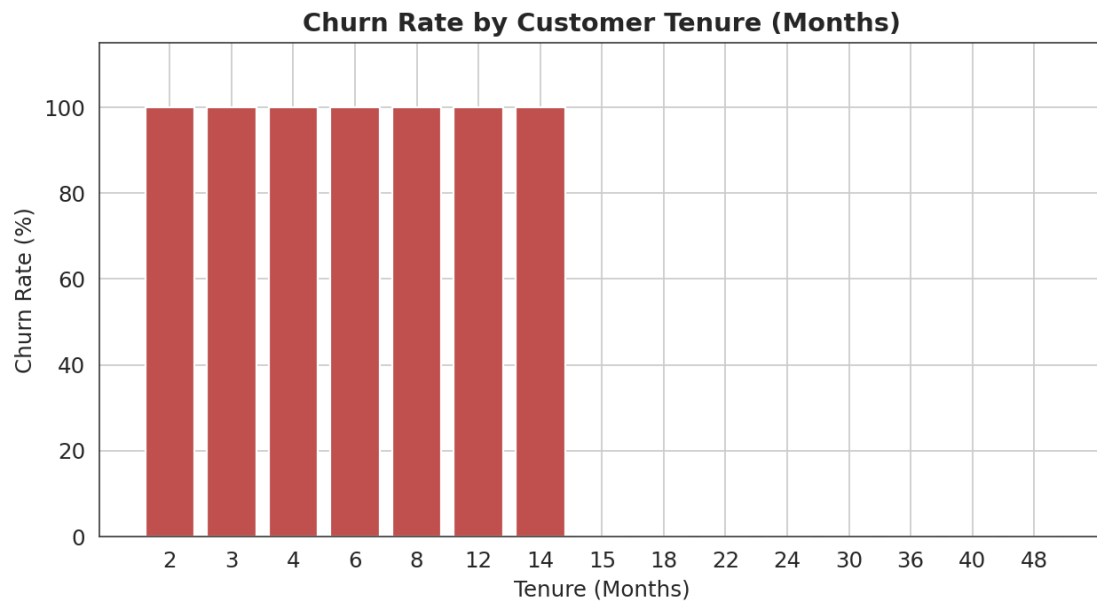


Figure 4: Churn rate across individual tenure values.

Every churned customer in the dataset had a tenure of 14 months or less; every customer with 15+ months of tenure was retained. Churn risk is heavily concentrated in the early subscription lifecycle.

5. Behavioral Drivers of Churn

5.1 Monthly Charges vs. Churn

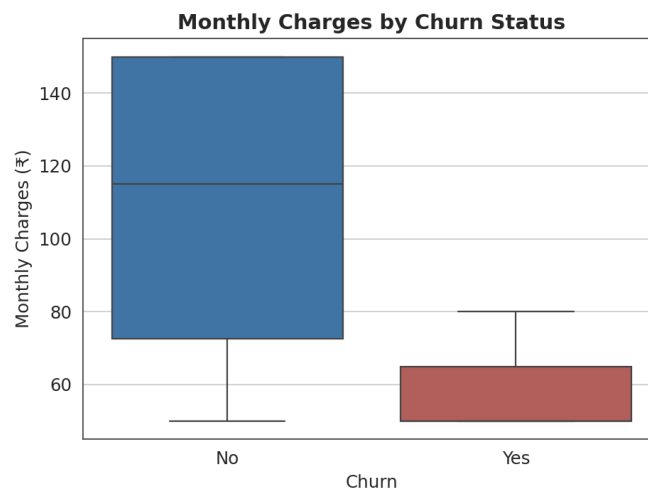


Figure 5: Distribution of monthly charges by churn status.

Churned customers pay a lower average monthly charge (Rs. 58.56) than retained customers (Rs. 107.49) — consistent with churn being concentrated in the lower-priced Basic tier and short Month-to-Month contracts.

5.2 Tenure vs. Churn

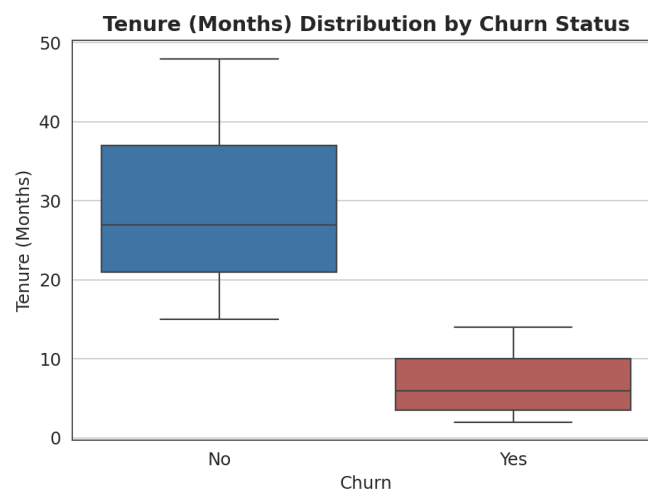


Figure 6: Distribution of account tenure by churn status.

Churned customers have an average tenure of just **7.0 months**, versus **29.1 months** for retained customers — a gap of over 22 months. Retention risk is front-loaded in the first year.

5.3 Correlation Analysis

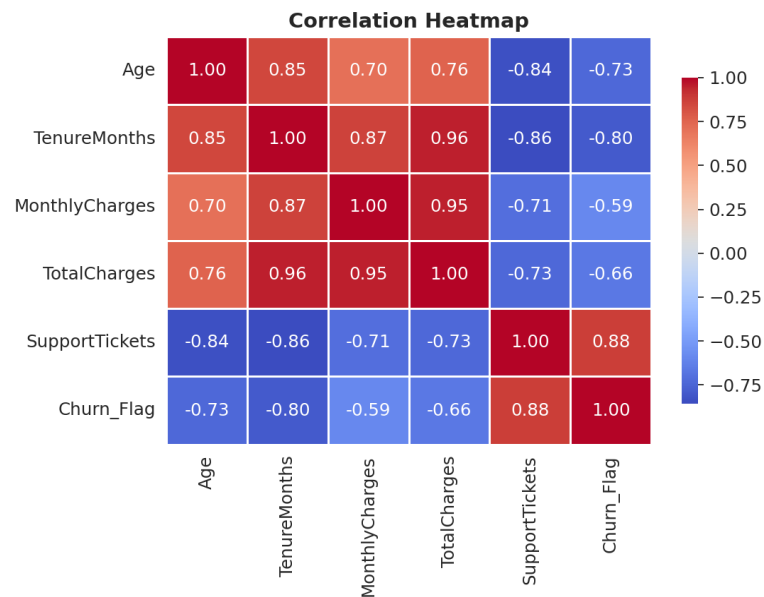


Figure 7: Correlation heatmap across numeric features and churn flag.

Support Tickets shows the strongest positive correlation with churn (**r = 0.88**), while Tenure (**r = -0.80**) and Age (**r = -0.73**) show the strongest negative correlations. More support tickets and less time on the platform both track closely with cancellation.

5.4 Driver Summary Table

Factor	Highest-Risk Group	Churn Rate
Contract Type	Month-to-Month	100.00%
Payment Method	Debit Card	100.00%
Support Tickets	5 tickets	100.00%
Subscription Type	Basic	71.43%

Supporting averages — **Monthly Charges:** Retained Rs. 107.49 vs. Churned Rs. 58.56 | **Tenure:** Retained 29.12 mo vs. Churned 7.00 mo | **Age:** Retained 46.62 vs. Churned 30.43 years.

6. Actionable Business Recommendations

1. Convert Month-to-Month customers to annual plans

Month-to-Month churn is 100% vs. 0% for annual contracts. Offer a discount (e.g., 1 month free) or bundled benefit for switching to a One Year plan, targeted at customers in their first 90 days.

2. Launch a proactive early-tenure retention program

All churn occurs within the first 14 months, and risk is highest in months 1–8. Trigger onboarding check-ins, usage nudges, and a dedicated success-manager outreach at 30, 60, and 90 days.

3. Flag and prioritize high support-ticket customers

Customers with 3+ support tickets churn at 100% (correlation $r = 0.88$). Route customers who cross this ticket threshold to a senior support tier or account manager before the issue drives cancellation.

4. Investigate Debit Card and UPI renewal friction

Debit Card (100%) and UPI (75%) users churn far more than Credit Card (33%) or Bank Transfer (0%) users. Audit these payment rails for failed auto-renewals and consider prompting a switch to Bank Transfer or Credit Card for long-term plans.

5. Strengthen the Basic tier value proposition

Basic-tier churn (71.4%) is far higher than Pro (50%) or Enterprise (0%). Introduce a light onboarding sequence and clearer upgrade path from Basic to Pro to increase perceived value early.

6. Track CLV-to-CAC alongside churn

With an estimated CLV of Rs. 1,591.55 and Rs. 409.93 in MRR already at risk, prioritize retention spend on segments identified above where the cost of intervention is well below the CLV recovered.

7. Conclusion

The analysis shows churn in this customer base is not random — it is heavily concentrated among Month-to-Month subscribers, customers within their first 14 months, and those generating multiple support tickets. Addressing contract structure, early-tenure engagement, and support escalation paths directly targets the segments responsible for the bulk of the 46.67% churn rate and the Rs. 409.93 in monthly recurring revenue currently at risk.

Note: This dataset contains 15 sample records used for methodology demonstration. The same formulas and segmentation approach should be re-applied to the full production dataset for operational decision-making.